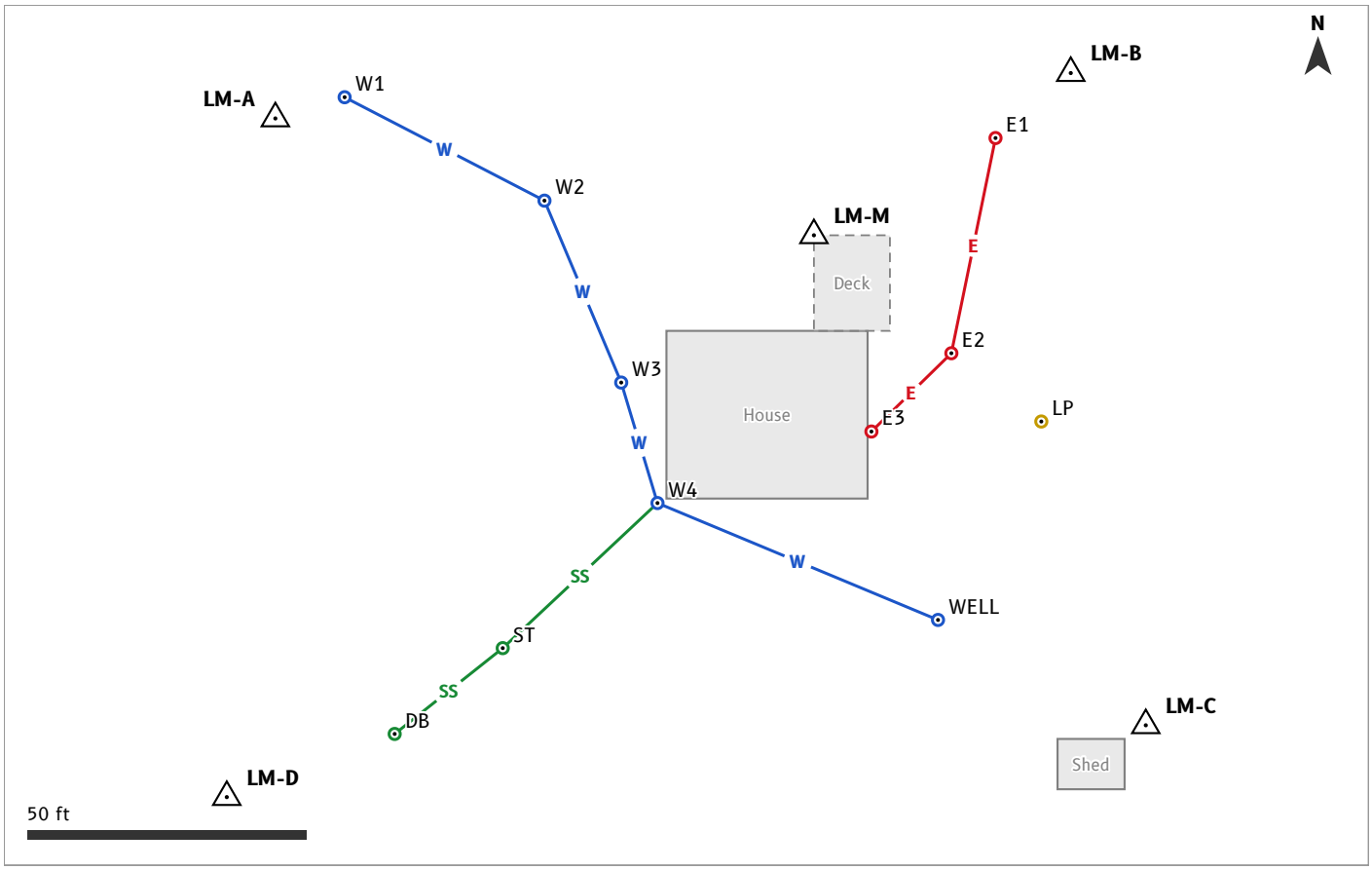
Example property

1234 Example Street

Surveyed by D. Panzino · Report printed 2026-10-16



Datum and solution

Coordinates	Local, in feet. LM-A held at N 5000.00, E 5000.00.
Basis of bearings	LM-A to LM-B = N 86°44'09" E (grid north).
Measurements	40 angles, 2 distances. Reading precision assumed 20" (one pointing); distances ±(0.02 ft + 1:5000).
Solution	16 of 16 located. 13 redundant measurements; readings fit at 0.34× the assumed precision.
Georeference	Grid north is 2°41'27" west of true north (±5°47'21", 95%), from a fit to the latitude/longitude of 4 landmarks; they agree with the survey to 6.6 ft RMS. Latitudes/longitudes below are good to a few feet, enough to walk to a spot with a phone.

Landmarks (control)

Name	Northing	Easting	±95%	Latitude, longitude	Description / how to find it
LM-A	5000.00	5000.00	0.00	40.772792, -73.975528	1/2" rebar with yellow cap, 2 ft E of driveway apron, NW corner
LM-B	5008.12	5142.37	0.08	40.772833, -73.975016	1/2" rebar with yellow cap at NE fence corner post
LM-C	4891.44	5155.80	0.11	40.772514, -73.974947	Chiseled X on concrete pad of shed, NE corner
LM-D	4878.59	4991.30	0.09	40.772458, -73.975538	1/2" rebar with yellow cap, SW corner by garden
LM-M	4979.10	5096.39	0.06	40.772747, -73.975176	16d nail in top of deck post, NW post of rear deck

Points

Name	Type	Northing	Easting	±95%	Check	Depth	Description

W1	Water	5003.80	5012.40	0.04	yes	42 in	Water meter box at street
W2	Water	4985.30	5048.17	0.05	yes	42 in	Water line bend (45 deg)
W3	Water	4952.75	5061.90	0.07	no	36 in	Water line tee to hose bib
W4	Water	4931.20	5068.40	0.08	no	36 in	Water entry at house foundation, SW corner
E1	Electric	4996.50	5128.90	0.08	yes	30 in	Transformer pedestal
E2	Electric	4958.00	5121.00	0.08	yes	24 in	Conduit bend
E3	Electric	4943.99	5106.69	0.07	no	24 in	House meter base on east wall, conduit up
ST	Sewer / septic	4905.24	5040.70	0.08	yes	18 in to lid	Septic tank inlet lid
DB	Sewer / septic	4889.89	5021.35	0.09	no	20 in	Septic distribution box
WELL	Water	4910.29	5118.60	0.09	yes		Well head (casing)
LP	Gas / oil	4945.79	5137.10	0.09	yes		Propane tank fill valve

“Check” = the point is seen from enough different directions that a misread angle would have shown up as a flagged reading.

Lines

Name	Type	Route	Length	Depth	Description
Water service	Water	W1 → W2 → W3 → W4	98.1 ft	36-42 in	1" PE
Electric service	Electric	E1 → E2 → E3	59.3 ft	24-30 in	2" PVC, 200A
Septic	Sewer / septic	W4 → ST → DB	62.7 ft		4" PVC
Well line	Water	WELL → W4	54.4 ft	48 in	1" PE

Obstacles

Name	Kind	Blocks sight	Corners	Description
House	Building	yes	4	Two storey, built 1978
Deck	Shed / garage / deck	no	4	Rear deck, 3 ft high, open below the rail
Shed	Shed / garage / deck	yes	4	10×12 shed on the concrete pad

Field book

Date	Station	Backsight	Target	Reading	Angle right	Distance	Residual	Note
2026-10-10	**LM-A**	LM-B @ 0 00 00	LM-D	97 21 40	97°21'40"		2"	
			LM-M	15 30 00	15°30'00"		1"	
			W1	346 13 40	346°13'40"		-1"	
			W2	20 14 20	20°14'20"		-1"	
			W3	40 37 00	40°37'00"		0"	
			W4	48 26 00	48°26'00"		-5"	
			E1	4 49 20	4°49'20"		9"	
			E2	22 24 20	22°24'20"		-8"	
			ST	70 01 20	70°01'20"		6"	
			DB	82 17 20	82°17'20"		1"	
2026-10-11	**LM-B**	LM-A @ 0 00 00	LM-C	266 42 00	266°42'00"		2"	
			LM-M	331 00 00	331°00'00"		0"	
			W2	349 38 40	349°38'40"		-3"	
			E1	322 29 00	322°29'00"		1"	
			E2	296 21 20	296°21'20"		-3"	
			E3	302 21 20	302°21'20"		0"	
			WELL	286 55 00	286°55'00"		1"	
			LP	278 05 40	278°05'40"		-6"	
2026-10-12	**LM-C**	LM-B @ 0 00 00	LM-D	272 05 40	272°05'40"		-12"	

			LM-M	332 26 20	332°26'20"		-5"	
			W4	301 01 20	301°01'20"		-6"	
			E1	352 12 20	352°12'20"		6"	
			E2	338 57 40	338°57'40"		-6"	
			E3	323 30 20	323°30'20"		0"	
			ST	283 24 20	283°24'20"		12"	
			DB	275 54 20	275°54'20"		4"	
			WELL	303 26 20	303°26'20"		0"	
2026-10-13	**LM-D**	LM-A @ 0 00 00	LM-C	81 26 20	81°26'20"		8"	
			W1	5 28 00	5°28'00"		6"	
			W2	23 57 20	23°57'20"		-2"	
			W3	39 29 40	39°29'40"		0"	
			W4	51 35 40	51°35'40"		-2"	
			ST	57 33 20	57°33'20"		5"	
			DB	65 17 40	65°17'40"		1"	
			WELL	71 55 20	71°55'20"		2"	
			LP	61 09 20	61°09'20"		-13"	
2026-10-14	**LM-M**	LM-A @ 0 00 00	LM-B	135 30 00	135°30'00"		-1"	
			LM-C	223 38 20	223°38'20"		-4"	
			W1	4 09 00	4°09'00"		-7"	
			LP	207 03 40	207°03'40"		5"	
	Taped: LM-A – LM-B					142.61	0.011 ft	Steel tape, 2 measurements averaged
	Taped: LM-C – LM-D					164.98	-0.011 ft	Steel tape

How to find each point

W1 — Water meter box at street N 5003.80 E 5012.40 · 40.772804, -73.975484

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **346°13'41"**; the point is **12.96 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **5°27'54"**; the point is **126.97 ft** out along that line.

Where the two lines cross: expected within ±0.05 ft (95%). 1 other setup is blocked by obstacles. Tape check: 87.55 ft from LM-M and 126.97 ft from LM-D; 12.96 ft from LM-A.

W2 — Water line bend (45 deg) N 4985.30 E 5048.17 · 40.772758, -73.975351

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **20°14'21"**; the point is **50.36 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **23°57'22"**; the point is **120.91 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). 1 other setup is blocked by obstacles. Tape check: 50.36 ft from LM-A and 120.91 ft from LM-D; 48.62 ft from LM-M.

W3 — Water line tee to hose bib N 4952.75 E 5061.90 · 40.772670, -73.975296

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **40°37'00"**; the point is **77.88 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **39°29'40"**; the point is **102.39 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). 3 other setups are blocked by obstacles. Tape check: 77.88 ft from LM-A and 102.39 ft from LM-D.

W4 — Water entry at house foundation, SW corner N 4931.20 E 5068.40 · 40.772612, -73.975269

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **48°26'05"**; the point is **97.02 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **51°35'42"**; the point is **93.33 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). 2 other setups are blocked by obstacles. Tape check: 93.33 ft from LM-D and 97.02 ft from LM-A; 96.02 ft from LM-C.

E1 — Transformer pedestal N 4996.50 E 5128.90 · 40.772799, -73.975062

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **322°28'59"**; the point is **17.79 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **352°12'14"**; the point is **108.45 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). 1 other setup is blocked by obstacles. Tape check: 36.87 ft from LM-M and 108.45 ft from LM-C; 17.79 ft from LM-B.

E2 — Conduit bend N 4958.00 E 5121.00 · 40.772692, -73.975084

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **296°21'23"**; the point is **54.49 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **208°22'43"**; the point is **32.41 ft** out along that line. *(line of sight not yet checked)*

Where the two lines cross: expected within ±0.02 ft (95%). 1 other setup is blocked by obstacles. Tape check: 32.41 ft from LM-M and 54.49 ft from LM-B; 75.10 ft from LM-C.

E3 — House meter base on east wall, conduit up N 4943.99 E 5106.69 · 40.772652, -73.975133

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **302°21'20"**; the point is **73.38 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **323°30'20"**; the point is **71.93 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). 3 other setups are blocked by obstacles. Tape check: 71.93 ft from LM-C and 73.38 ft from LM-B.

ST — Septic tank inlet lid N 4905.24 E 5040.70 · 40.772537, -73.975365

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **70°01'14"**; the point is **103.13 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **57°33'15"**; the point is **56.13 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). 2 other setups are blocked by obstacles. Tape check: 56.13 ft from LM-D and 103.13 ft from LM-A; 115.92 ft from LM-C.

DB — Septic distribution box N 4889.89 E 5021.35 · 40.772493, -73.975432

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **82°17'19"**; the point is **112.16 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **65°17'39"**; the point is **32.10 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). 2 other setups are blocked by obstacles. Tape check: 32.10 ft from LM-D and 112.16 ft from LM-A; 134.45 ft from LM-C.

WELL — Well head (casing) N 4910.29 E 5118.60 · 40.772561, -73.975085

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **286°54'59"**; the point is **100.68 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **303°26'20"**; the point is **41.69 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). 2 other setups are blocked by obstacles. Tape check: 41.69 ft from LM-C and 100.68 ft from LM-B; 131.19 ft from LM-D.

LP — Propane tank fill valve N 4945.79 E 5137.10 · 40.772661, -73.975024

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **278°05'46"**; the point is **62.56 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **207°03'35"**; the point is **52.60 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). 1 other setup is blocked by obstacles. Tape check: 52.60 ft from LM-M and 62.56 ft from LM-B; 57.47 ft from LM-C.

Notes

1. Coordinates are local (assumed). Bearings are referenced to the line LM-A to LM-B.
2. Depths are to top of pipe/conduit, measured when the trench was open.